

HDOC Day-42

Menu-Driven C Program – String Operations

Aim

To write a **menu-driven C program** using separate user-defined functions to:

1. Find the **longest word in a sentence**.
2. Check whether one string is a **rotation of another string**.
3. Exit the program.

C Program

```
#include <stdio.h>
#include <string.h>

void longestWord()
{
    char str[200], word[100], longest[100];
    int i = 0, j = 0, max = 0;

    printf("Enter a sentence: ");
    getchar();
    fgets(str, sizeof(str), stdin);

    while (1)
    {
        if (str[i] != ' ' && str[i] != '\n' && str[i] != '\0')
        {
            word[j++] = str[i];
        }
        else
        {
            word[j] = '\0';

            if (j > max)
            {
                max = j;
                strcpy(longest, word);
            }

            j = 0;

            if (str[i] == '\0' || str[i] == '\n')
                break;
        }
        i++;
    }
}
```

```

        printf("Longest word: %s\n", longest);
        printf("Length: %d\n", max);
    }

void checkRotation()
{
    char str1[100], str2[100], temp[200];

    printf("Enter first string: ");
    scanf("%s", str1);

    printf("Enter second string: ");
    scanf("%s", str2);

    if (strlen(str1) != strlen(str2))
    {
        printf("The strings are NOT rotations of each other.\n");
        return;
    }

    strcpy(temp, str1);
    strcat(temp, str1);

    if (strstr(temp, str2) != NULL)
        printf("The strings are rotations of each other.\n");
    else
        printf("The strings are NOT rotations of each other.\n");
}

int main()
{
    int choice;

    do
    {
        printf("\n----- MENU ----- \n");
        printf("1. Find longest word in a sentence\n");
        printf("2. Check rotation of two strings\n");
        printf("3. Exit\n");

        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch (choice)
        {
            case 1:
                longestWord();
                break;

            case 2:
                checkRotation();
                break;

            case 3:
                printf("Program terminated.\n");
                break;
        }
    }

```

```

        default:
            printf("Invalid choice! Please try again.\n");
        }

    } while (choice != 3);

    return 0;
}

```

Sample Output 1 — Longest Word

```

----- MENU -----
1. Find longest word in a sentence
2. Check rotation of two strings
3. Exit
Enter your choice: 1

Enter a sentence: Programming in C is interesting
Longest word: Programming
Length: 11

```

Sample Output 2 — String Rotation

```

----- MENU -----
1. Find longest word in a sentence
2. Check rotation of two strings
3. Exit
Enter your choice: 2

Enter first string: ABCD
Enter second string: CDAB
The strings are rotations of each other.

```

Logic for Rotation

Suppose:

```

String 1 = ABCD
String 2 = CDAB

```

Concatenate String 1 with itself:

```

ABCD + ABCD = ABCDABCD

```

Since CDAB occurs inside ABCDABCD, **CDAB is a rotation of ABCD.**

Functions Used

Function	Purpose
longestWord()	Finds the longest word in the entered sentence

Function	Purpose
<code>checkRotation()</code>	Checks whether the second string is a rotation of the first
<code>strlen()</code>	Finds string length
<code>strcpy()</code>	Copies one string into another
<code>strcat()</code>	Concatenates two strings
<code>strstr()</code>	Searches for one string inside another

Result

Thus, the menu-driven C program was successfully implemented to **find the longest word in a sentence** and **check whether two strings are rotations of each other**.